

Irfan Sudarani

+91 6361850072 — irfanrs024@gmail.com — linkedin.com/in/irfanrs
github.com/Irfan-RS — codolio.com/profile/irfan72 — irfansudarani.netlify.app

Summary

Seeking a Software Development Intern role to contribute to building scalable and impactful systems, with a strong focus on problem-solving in Full-Stack Web Development. Experienced in developing APIs, working with databases, and building backend systems, with interest in system design and production-ready applications.

Education

B.M.S. College of Engineering, Bengaluru <i>B.E. in Computer Science and Engineering</i>	2023 – 2027 CGPA: 8.57
Smt. Vidya P Hanchimani PU College, Dharwad <i>Pre-University Course (PCMC)</i>	2021 – 2023 KCET Rank: 2674

Skills

Languages: Java, Python, C, JavaScript

AIML: NumPy, Pandas, BeautifulSoup, Matplotlib, Scikit-learn, TensorFlow

Databases: MongoDB, Cassandra, MySQL, PostgreSQL

Core: DBMS, OOPs, Operating Systems, Computer Networks, System Design, Basic Cloud Technologies.

Tools: Git, GitHub, VS Code, Docker, Postman

OS: Windows, Ubuntu

Projects

Campus Space

[GitHub](#)

Technologies: React.js, tailwind CSS, Node.js, Express.js, MongoDB (Atlas), AWS S3, Firebase Auth

- Refactored a monolithic Node.js/Express backend into a highly modular, layered architecture (separating Config, Models, Controllers, Routes, and Middleware), reducing code coupling by 80% and significantly improving maintainability and readability.
- Integrated AWS S3 for media storage and implemented an automated asset lifecycle workflow, deleting orphaned profile images from the S3 bucket upon new uploads to prevent storage leaks and reduce cloud hosting costs.
- Designed and implemented a dynamic user dashboard and a paginated notification system (tracking likes, comments, and threaded replies) utilizing MongoDB/Mongoose indexing and population to optimize query performance and ensure sub-second response times.

AI-Powered Online Proctoring System

[GitHub](#)

Technologies: React.js, TypeScript, FastAPI, SQLite, OpenCV, MediaPipe, YOLOv8

- Engineered a real-time computer vision pipeline by integrating **MediaPipe Face Mesh**, **InsightFace**, and **YOLOv8** with **OpenCV** to perform multi-face tracking, head-pose estimation (Pitch/Yaw/Roll), off-axis gaze tracking, impersonation verification, and electronic device detection.
- Designed a high-throughput API backend in **FastAPI** utilizing **SQLite** optimized with **Write-Ahead Logging (WAL)** and connection retry timeouts to support concurrent read/write operations from multiple simultaneous student webcam streams.
- Developed a responsive student dashboard and admin monitoring panel in **React.js**, **Vite**, and **TypeScript** using **WebSockets** for real-time telemetry updates and debounced browser event listeners to track tab-switching and mouse boundary violations.

Leadership

Senior Coordinator – NSS Unit, BMSCE

Oct 2023 – Present

- Represented VTU and Karnataka at the National Integration Camp (NIC) at the national level.
- Planned and executed a 7-day residential NSS camp at a government school, managing logistics, scheduling, and on-ground coordination for two consecutive years.
- Organized and coordinated 50+ NSS events ensuring smooth execution and active student participation.